

A Technical Introduction to Algorithmic bias

Section 1

Gender discrimination In 2016, the professional networking site LinkedIn was discovered to recommend male variations of women's names in response to search queries. The site did not make similar recommendations in searches for men's names. For example, "Andrea" would bring up a prompt asking if users meant "Andrew", but queries for "Andrew" did not ask if users meant to find "Andrea". The company said this was the result of an analysis of users' interactions with the site. In 2012, the department store franchise Target was cited for gathering data points to infer when female customers were pregnant, even if they had not announced it, and then sharing that information with marketing partners. Because the data had been predicted, rather than directly observed or reported, the company had no legal obligation to protect the privacy of those customers. Web search algorithms have also been accused of bias. Google's results may prioritize pornographic content in search terms related to sexuality, for example, "lesbian". This bias extends to the search engine showing popular but sexualized content in neutral searches. For example, "Top 25 Sexiest Women Athletes" articles displayed as first-page results in searches for "women athletes". In 2017, Google adjusted these results along with others that surfaced hate groups, racist views, child abuse and pornography, and other upsetting and offensive content. Other examples include the display of higher-paying jobs to male applicants on job search websites. Researchers have also identified that machine translation exhibits a strong tendency towards male defaults. In particular, this is observed in fields linked to unbalanced gender distribution, including STEM occupations. In fact, current machine translation systems fail to reproduce the real world distribution of female workers. In 2015, Amazon.com turned off an AI system it developed to screen job applications when they realized it was biased against women. The recruitment tool excluded applicants who attended all-women's colleges and resumes that included the word "women's". A similar problem emerged with music streaming services—In 2019, it was discovered that the recommender system algorithm used by Spotify was biased against female artists. Spotify's song recommendations suggested more male artists over female artists.

Section 2

There have been several attempts to create methods and tools that can detect and observe biases within an algorithm. These emergent fields focus on tools which are typically applied to the (training) data used by the program rather than the algorithm's internal processes. These

methods may also analyze a program's output and its usefulness and therefore may involve the analysis of its confusion matrix (or table of confusion). Explainable AI to detect algorithm Bias is a suggested way to detect the existence of bias in an algorithm or learning model. Using machine learning to detect bias is called, "conducting an AI audit", where the "auditor" is an algorithm that goes through the AI model and the training data to identify biases. Ensuring that an AI tool such as a classifier is free from bias is more difficult than just removing the sensitive information from its input signals, because this is typically implicit in other signals. For example, the hobbies, sports and schools attended by a job candidate might reveal their gender to the software, even when this is removed from the analysis. Solutions to this problem involve ensuring that the intelligent agent does not have any information that could be used to reconstruct the protected and sensitive information about the subject, as first demonstrated in where a deep learning network was simultaneously trained to learn a task while at the same time being completely agnostic about the protected feature. A simpler method was proposed in the context of word embeddings, and involves removing information that is correlated with the protected characteristic. Currently, a new IEEE standard is being drafted that aims to specify methodologies which help creators of algorithms eliminate issues of bias and articulate transparency (i.e. to authorities or end users) about the function and possible effects of their algorithms. The project was approved February 2017 and is sponsored by the Software & Systems Engineering Standards Committee, a committee chartered by the IEEE Computer Society. A draft of the standard is expected to be submitted for balloting in June 2019. The standard was published in January 2025. In 2022, the IEEE released a standard aimed at specifying methodologies to help creators of algorithms address issues of bias and promote transparency regarding the function and potential effects of their algorithms. The project, initially approved in February 2017, was sponsored by the Software & Systems Engineering Standards Committee, a committee under the IEEE Computer Society. The standard provides guidelines for articulating transparency to authorities or end users and mitigating algorithmic biases.

Section 3

Interdisciplinarity and Collaboration Integrating interdisciplinarity and collaboration in developing of AI systems can play a critical role in tackling algorithmic bias. Integrating insights, expertise, and perspectives from disciplines outside of computer science can foster a better understanding of the impact data driven solutions have on society. An example of this in AI research is PACT or Participatory Approach to enable Capabilities in communiTies, a proposed framework for facilitating collaboration when developing AI driven solutions concerned with social impact. This framework identifies

guiding principals for stakeholder participation when working on AI for Social Good (AI4SG) projects. PACT attempts to reify the importance of decolonizing and power-shifting efforts in the design of human-centered AI solutions. An academic initiative in this regard is the Stanford University's Institute for Human-Centered Artificial Intelligence which aims to foster multidisciplinary collaboration. The mission of the institute is to advance artificial intelligence (AI) research, education, policy and practice to improve the human condition. Collaboration with outside experts and various stakeholders facilitates ethical, inclusive, and accountable development of intelligent systems. It incorporates ethical considerations, understands the social and cultural context, promotes human-centered design, leverages technical expertise, and addresses policy and legal considerations. Collaboration across disciplines is essential to effectively mitigate bias in AI systems and ensure that AI technologies are fair, transparent, and accountable.

Section 4

Selection bias Selection bias refers to the inherent tendency of large language models to favor certain option identifiers irrespective of the actual content of the options. This bias primarily stems from token bias—that is, the model assigns a higher a priori probability to specific answer tokens (such as "A") when generating responses. As a result, when the ordering of options is altered (for example, by systematically moving the correct answer to different positions), the model's performance can fluctuate significantly. This phenomenon undermines the reliability of large language models in multiple-choice settings.